Skin Lesion Classification Case Study Rubric

DS 4002

Due Date: May 6

Submission Format: Upload GitHub Link to Canvas

Individual Assignment

Why am I doing this? This case study introduces students to machine learning approaches for classifying skin lesions using image data and metadata. The goal is to help students understand how modeling decisions impact performance and how real-world challenges, such as class imbalance, affect results. This project also encourages students to think about how data science can be applied in healthcare to improve accessibility and early detection.

What am I going to do? In this case study, you will explore how machine learning models can be used to classify pigmented skin lesions using dermatoscopic images and patient metadata. The GitHub repository for this case study can be found at https://github.com/katherinejlai/skin-lesions-case-study. The GitHub repository includes:

- Two example model scripts:
    - ResNet50-based model
    - MobileNetV2-based model
- Implementations of both:
    - Image-only model
    - Multimodal model

Using these resources, you will:

- Run and evaluate the provided models
- Compare the performance of image-only vs. Multimodal approaches
- Analyze metrics such as F1, precision, recall, and confusion matrices

You are encouraged to extend this work by:

- Experimenting with different pretrained models
- Modifying the model architecture
- Improving performance through tuning or feature selection

Tips for Success:

- Carefully review the provided background materials to understand both how dermatologists identify skin lesions and how machine learning models classify image data.

- Start by running the provided scripts before making any changes, so you understand the baseline performance of each model.
- When experimenting with improvements, make small, controlled changes so you can clearly identify what impacts performance.

How will I know I have succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Spec Category	Spec Details

Formatting	• One GitHub repository containing all materials:
	  o README.md (auto displays)
	  o LICENSE.md (use MIT as default)
	  o Data folder
	  o Scripts folder
	  o Output folder
	  o Reflection pdf

README.md	• Goal: Provide a clear overview of the case study and repository • List software and packages used for analysis • Explain contents within folders using documentation map • Explain how results can be replicated • Include references
LICENSE.md	• Goal: Define how others can use and cite the repository • Use the MIT license here
Data Folder	• Goal: This folder contains all the data for the project • Include the metadata file and provide a link to the image dataset due to size
Scripts Folder	• Goal: This folder will contain all source code used in the project • This should include the edited version of the example models or the code of a new pretrained model
Output Folder	• Goal: This folder should showcase model results and performance • Include evaluation metrics • And at least one visualization (e.g., confusion matrix)
Reflection	• Goal: Analyze results and connect them to real-world implications • Discuss model performance and limitations • Address any biases in the dataset • Compare image-only and multimodal approaches • Compare performance across different pre-trained models

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